STAT 443/851 Theory of Linear Models (University of Saskatchewan, 2026-01)

Description

This course is a rigorous examination of the general linear models using vector space theory, in particular the approach regarding least square as projection. The topics include: vector space; projection; matrix algebra; generalized inverses; quadratic forms; theory for point estimation; theory for hypothesis test; theory for non-full-rank models.

Prerequisite(s): MATH 164 (formerly MATH 264) or MATH 266, STAT 342, and STAT 344 or 345.

Instructor

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Times and Places

Lecture Classroom: MCLN 242.1, MWF 9:30-10:20; Office hour: TBA; Lab: no lab.

Course Materials

- See the course page, which is the primary source for learning.
- Recommended reading: LINEAR MODELS IN STATISTICS, Second Edition, by Alvin C. Rencher and G. Bruce Schaalje, ISBN 978-0-471-75498-5 (cloth).

Learning Outcomes

After completing this course, students are expected to grasp the following knowledge and skills:

Table 1

Topic	Learning Outcomes (Knowledge & Skills)	Perc
Vector Space	**Understanding** the geometric interpretation of subspaces and the relationship between the Hat matrix (H) and the Partial Residual matrix. **Numerically and symbolically calculating** orthogonal projections onto these spaces.	15%
Matrix Algebra	**Understanding** advanced matrix theory, including spectral decomposition and idempotent properties. **Deriving** properties of projection matrices and **calculating** generalized inverses.	15%
Multivariate Normal	**Understanding** the properties of multivariate normal vectors and correlation coefficients. **Deriving** the mean and covariance of transformed normal vectors and **calculating** conditional expectations.	15%
Quadratic Forms	**Understanding** the distributions of quadratic forms. **Verifying** independence of quadratic forms and **applying** Cochran's Theorem to derive distributions and degrees of freedom.	15%
Multiple Regression	**Understanding** the general linear hypothesis. **Calculating** LS estimators, error variance, and adjusted R^2 with CI. **Implementing** F-tests and t-tests; **constructing** confidence/prediction regions; and **symbolically deriving** SSH, SSE, and the F-distribution.	25%
Non-full-rank Models	**Understanding** identifiability and estimable functions in over-parameterized models. **Identifying** estimable linear combinations and **formulating** valid hypothesis tests in ANOVA frameworks.	15%

Tentative Schedule

Table 2

Date (Mon)	Week	Lecture Topic	Tests & Notes
Jan 05	1	1. Vector Space and Projection	Jan 07: First day of class
Jan 12	2	1. Vector Space and Projection	
Jan 19	3	2. Matrix Algebra	**Jan 25: Assignment 1 due**
Jan 26	4	2. Matrix Algebra	
Feb 02	5	3. Distribution of Multivariate Normal	
Feb 09	6	4. Distribution of Quadratic Forms	
Feb 16	N/A	—	**Reading Week – No classes**
Feb 23	7	4. Distribution of Quadratic Forms	**Mar 01: Assignment 2 due**
Mar 02	8	5. Theory for Multiple Regression	**Mar 06: Midterm (during class)**
Mar 09	9	5. Theory for Multiple Regression	
Mar 16	10	5. Theory for Multiple Regression	
Mar 23	11	6. Non-full-rank Models	
Mar 30	12	6. Non-full-rank Models	**Apr 05: Assignment 3 due**
Apr 06	13	Review	Apr 06: Last lecture

! Important

The schedule may change depending on the course pace. The exact assignment and test dates are given on Canvas.

Evaluation

Grading Scheme

3 assignments: 3x10% = 30%, 1 term test: 20%, final exam: 50%.

Assignments and Tests

Assignment questions are released in the one-drive folder. You will submit your solutions via Canvas. Undergraduate students will be assigned different assignments and tests.

Assignments

- I will accept late assignments only for three (3) days beyond the due date. The penalty is 10 percentage points per day.

- Answer the questions in the order they appear. Neatness is important.
- Solutions to problems are to be included. Simple answers without work will receive few (or no!) marks.
- Submitting: Save your assignment as **one PDF file** and upload to Canvas.

Midterm

- The midterm is given in the class period.
- Type: Short-answer questions, problem-solving, open-book.
- Calculator: A scientific calculator is allowed.

Final Exam

- The final exam will cover material of the entire course.
- Length: 3-hour in-person exam.
- Type: Short-answer questions, problem-solving, open-book.

Pass Criteria

The **final exam is a required component**. Students must complete the final exam to be eligible for a passing grade.

Use of Generative AI and Electronic Devices

1. **AI for Learning:** Students are encouraged to use Generative AI tools as a study aid. However, **all submitted work must be your own.** Directly copying AI output and submitting it as your own may receive a **severe penalty**.

2. **Electronic Devices:** All tests are **Open Book (Paper-based)**.

- **No Electronic Devices:** Laptops, tablets, and smartwatches are **NOT allowed** during exams.
- **Phone Exception:** Smartphones must be stowed away. They may **only** be used at the end to take photos of answer sheets for submission.